Lesson 03 — Assistive and Adaptive Technology, Portable Devices, and the Environment

Strand 1 — Introduction to Computing · Sub-strand 2 — Technology in the Community Indicators: B8.1.2.1.1 (adaptive and assistive technologies) · B8.1.2.1.2 (portable computing devices) · B7.1.2.1.4 (environmental impact of computers) · B7.1.2.1.5 (environmentally responsible practices) Source coverage: BECE Mock 0201026/27 Computing (2026) P1 Q7, 8, 13 · P2 Q1(c)(i)–(iii) + authentic bank questions.

Preamble (why this lesson, in everyday words)

A computer should serve everybody — including students who cannot see, cannot hear, cannot walk, or cannot speak. Engineers have built special equipment so that no learner is left behind. That equipment has two names you must never mix up: assistive technology and adaptive technology.

The technology also packs itself smaller and smaller, so today a whole computer fits in your pocket — these are the portable devices, with their own strengths and one famous weakness (the battery!).

And because computers rule our lives, they also touch the environment: power use, old parts dumped in the soil, and short-lived gadgets. We finish the lesson by learning how to use technology responsibly.

Words to know first

Word	Simple meaning
Disability	A condition that limits a person's movement, senses or activity.
Assistive technology	Equipment that adds an ability the person was missing (e.g. a hearing aid adds hearing sound louder).
Adaptive technology	Equipment changed/arranged to fit how the person already works (e.g. a tablet app changed to picture-buttons for a non-verbal student).
Accessible	Able to be reached and used by everyone, including persons with disabilities.
Portable	Small and light enough to be carried easily.
Tactile	Felt by touch.
E-waste	Old, broken electronic parts thrown away.
Product life cycle	From manufacture → use → disposal of a device.

Memory hook: Assistive = Adds. Adaptive = Adjusted.

Teaching body — Section 1: Assistive technology (Theory Q&A)

Q1. What is assistive technology? Ans. Assistive technology is any device or equipment used to increase, maintain or improve the functional abilities of persons with disabilities.

Q2. Give four examples of assistive technology used in schools. Ans.

- Hearing aid — makes sounds louder for learners who are hard of hearing.
- Braille printers and Braille keyboards — give tactile output for visually impaired learners.
- Screen readers — read aloud the text displayed on a computer screen.
- Wheelchairs and prosthetic limbs — support movement for learners with physical disabilities.

Q3. What is a hearing aid? Ans. A hearing aid is a small electronic device worn behind the ear that makes sounds louder for a person who is hard of hearing.

Q4. What is a prosthetic limb? Ans. A prosthetic limb is an artificial limb that replaces a missing part of the body, such as a leg below the knee.

Q5. State one benefit of an artificial (prosthetic) leg. Ans. It allows the user to stand, walk and move about again, restoring independence.

Q6. What is a screen reader? Ans. A screen reader is software that reads out the text on the screen so a visually impaired person can use a computer by listening.

Q7. Why is a screen reader an example of assistive technology and not an ordinary keyboard? Ans. Because it is designed mainly to support a person with a disability (blindness), adding an ability that normal computer output does not give.

Teaching body — Section 2: Adaptive technology

Q8. What is adaptive technology? Ans. Adaptive technology is equipment modified or specially arranged so that it fits the way a person with a disability already works, making existing technology accessible.

Q9. Give three examples of adaptive technology. Ans.

- A tablet loaded with a picture-based communication app for a non-verbal student.
- A strap-on handle fixed on a spoon or fork so a person with a weak grip can hold it.
- A keyboard with raised dots so a visually impaired person can read keys by touch.

Q10. What is the main difference between assistive and adaptive technology? Ans. Assistive technology is a purpose-built device that adds a missing ability to the user, while adaptive technology is an adaptation of technology or everyday tools to make them usable by the person.

Point	Assistive	Adaptive
Purpose	Adds an ability the user lacks	Adjusts existing tools to fit the user
Examples	Hearing aid, screen reader, prosthetic leg, wheelchair	Picture-based tablet app, strap-on spoon handle, raised-dot keyboard
Who it "changes"	Replaces/extends a body ability	Changes the equipment, not the body

Q11. Why do schools procure assistive and adaptive equipment? Ans. To make classrooms more accessible, so that students living with disabilities can learn together with their mates.

Q12. How does adaptive technology support inclusive education? Ans. It lets learners with different disabilities use the same classroom, books and computers by adjusting the tools to each learner's needs.

Teaching body — Section 3: Portable computing devices

Q13. What is a portable computing device? Ans. A portable computing device is a small, light computer that can easily be carried and used in different places.

Q14. Give four examples of portable computing devices. Ans. Laptop, tablet, smartphone, and notebook. (Also: ultrabook, e-reader.)

Q15. State two advantages of portable computing devices. Ans. (Any two) Portability — they can be carried and used anywhere; they run on rechargeable batteries; they are compact in size; they support multitasking.

Q16. State one disadvantage of portable computing devices. Ans. Their batteries are limited, so the device must be charged often. (Also: smaller screens and keyboards; easier to lose or steal; weaker performance than desktops.)

Q17. How do portable computing devices affect our everyday lives? Ans. They let us learn, bank, communicate, navigate and entertain ourselves at any place and time without being tied to a desk.

Q18. Why is battery life the most common problem of portable devices? Ans. Because the device works on a small rechargeable battery which runs down quickly with screen use, Internet use and heavy apps.

Teaching body — Section 4: Computers and the environment

Q19. State the negative impacts of computers on the environment. Ans. (Any four)

- High power consumption when many computers are running or idling.
- Short product life cycle — devices are replaced quickly and dumped.
- E-waste pollution when old parts are thrown into the soil.
- Heat and noise from machines and data centres.
- Raw materials and chemicals used in manufacturing.

Q20. Which factor contributes most directly to the environmental impact of computers? Ans. The short product life cycle — computers replaced and discarded often create the most waste. (Low power consumption and efficient resource use are good practices, not impacts.)

Q21. What is e-waste? Ans. E-waste is discarded electrical or electronic equipment, such as dead computers, monitors and cables, which pollutes the soil and water.

Q22. Propose environmentally responsible practices for computer use. Ans.

- Switch off or sleep computers when not in use, to save power.
- Repair, upgrade and reuse old computers instead of dumping them.
- Send dead parts to certified e-waste recyclers.
- Buy devices with energy-saving labels and long warranties.
- Avoid printing unnecessarily, to save paper and ink.

Q23. What useful component can be created from disposed computer parts? Ans. A working component such as a small storage or display unit assembled from recycled parts, e.g. a media player from an old hard drive and monitor.

Worked Exam Exemplars — 15 solved examples

Example 1 [Mock 0201026/27, P1 Q8] Which of the following is an example of assistive technology? — Keyboard / Screen reader / Bluetooth speaker / Selfie stick. (1 mark) Ans. B — Screen reader. It reads the screen aloud for blind users; the others are ordinary tools anyone uses.

Example 2 [Mock 0201026/27, P1 Q7] Which of the following is a disadvantage of portable computing devices? — Portability / Limited battery / Multitasking / Compact size. (1 mark) Ans. B — Limited battery. Options A, C and D are advantages.

Example 3 [Mock 0201026/27, P1 Q13] Which of the following contributes to the environmental impact of computers? — Minimal power consumption / Short product life cycle / Efficient use of resources / Short manufacturing processes. (1 mark) Ans. B — Short product life cycle. Fast replacement and disposal of devices creates e-waste; the other options are eco-friendly practices.

Example 4 [Mock 0201026/27, P2 Q1(c)(i)] Name each device described: I: A small device worn behind the ear that makes sounds louder; II: A tablet with a picture-based app for a non-verbal student; III: A strap-on handle that helps someone with a weak grip hold a spoon or fork; IV: An artificial limb that replaces a leg below the knee; V: A chair that helps someone with limited mobility move around; VI: A keyboard with raised dots read by touch. (3 marks) Ans. (½ mark each)

- I: Hearing aid
- II: Picture communication tablet (AAC tablet)
- III: Adaptive strap-on handle (strap-on grip aid)
- IV: Prosthetic leg (prosthesis)
- V: Wheelchair
- VI: Braille keyboard

Example 5 [Mock 0201026/27, P2 Q1(c)(ii)] Classify each of the devices I to VI as assistive or adaptive technology. (3 marks, ½ each) Ans.

Device	Classification	Reason

I Hearing aid	Assistive	Purpose-built device that adds hearing ability
II Picture tablet	Adaptive	A normal tablet adapted with a picture app to fit the user
III Strap-on handle	Adaptive	An everyday spoon/fork adjusted to fit the user's grip
IV Prosthetic leg	Assistive	Adds mobility to the user
V Wheelchair	Assistive	Adds movement ability
VI Raised-dot keyboard	Adaptive	A normal keyboard adjusted, giving tactile keys

Example 6 [Mock 0201026/27, P2 Q1(c)(iii)] State one benefit of each of devices IV and V for the disabilities they support. (3 marks) Ans. IV (prosthetic leg): enables the user to walk, stand and move independently. V (wheelchair): enables a user with limited mobility to move around, attend school and work without needing to stand.

Example 7 [Own, Paper 2 style] Distinguish between assistive and adaptive technology, giving one example of each. (2–4 marks) Ans. Assistive technology is a purpose-built device that adds an ability the user lacks, e.g. a hearing aid; adaptive technology is existing technology modified to suit the user, e.g. a tablet fitted with a picture-communication app. (2 marks per correct contrast + example = up to 4)

Example 8 [Own, Paper 2 style] Mention three pieces of technology that make a classroom more accessible. (3 marks) Ans. (1 mark each) Screen reader; hearing aid; Braille materials (keyboard/printer); ramps and adjustable desks for wheelchair users.

Example 9 [Own, Paper 2 style] A visually impaired student must now take the same computer test as the rest of the class. Suggest two technology adaptations the school can provide. (4 marks, 2 each) Ans. (a) A screen reader, so the computer reads each question aloud; (b) a Braille keyboard/Braille printout of the questions, so he can read and type by touch.

Example 10 [Own, Bank style] State two advantages and one disadvantage of a smartphone as a computing device. (3 marks) Ans. Advantages: it is portable and works anywhere; it connects to the Internet for learning and communication. Disadvantage: limited battery life — it needs frequent charging. (1 mark each)

Example 11 [Own, Bank style] Kofi abandoned his old desktop in the storeroom and the family wants to dump it behind the house. Mention the environmental problem this will cause and give a better practice. (4 marks) Ans. Problem: it becomes e-waste — the metals and chemicals leak into the soil and water and harm people and animals. Better practice: send it to a certified e-waste recycler, or repair/donate it for reuse. (2 + 2)

Example 12 [Own, Bank style] State three environmentally responsible practices for using computers at school. (3 marks) Ans. (1 mark each) Switch off/idle-sleep computers after class; print only when necessary; recycle or donate old computers instead of dumping them.

Example 13 [Own, Paper 1 converted] Which of these is NOT a portable computing device? — Laptop / Tablet / Desktop PC / Notebook. Ans. Desktop PC — it is fixed on a desk and is not designed to be carried.

Example 14 [Own, Paper 2 style] Explain how a hearing aid improves learning for a hard-of-hearing student. (2 marks) Ans. It makes the teacher's voice louder, so the student hears instructions and joins class discussion with the rest of the class.

Example 15 [Own, Synthesis] A school IT lab of 20 computers runs all day even at break time. State two impacts this has and propose one practice to fix it. (4 marks) Ans. Impacts: high power consumption/wasted electricity; heat build-up that can shorten device life. Practice: switch computers off (or set sleep mode) when leaving for break, and switch off at close of day.

Exercises — 5 sections × 5 questions (no multiple choice)

Exercise 1 — Assistive technology

1. Define assistive technology. (2 marks)
2. Give three examples of assistive technology used in schools. (3 marks)
3. What is a hearing aid used for? (2 marks)
4. Name the device that produces raised dots on paper read by the fingertips. (1 mark)
5. State one benefit of a prosthetic limb for a learner who lost his leg. (2 marks)

Exercise 2 — Adaptive technology and classification

1. Define adaptive technology. (2 marks)
2. Give two examples of adaptive technology. (2 marks)
3. A non-verbal student uses a tablet app showing picture buttons to build sentences. Name the adaptation and classify it. (2 marks)
4. Copy and complete the table: Device → Assistive or Adaptive. Devices: hearing aid; strap-on spoon handle; screen reader; raised-dot keyboard. (4 marks)
5. State one way adaptive technology supports inclusive education. (2 marks)

Exercise 3 — Portable devices

1. What is a portable computing device? (2 marks)
2. Give four examples of portable computing devices. (2 marks)
3. State two advantages of portable computing devices. (2 marks)
4. State two disadvantages of portable computing devices. (2 marks)
5. Ama finished a group project at her aunt's house using her tablet and sent it to the class group. Which features of the tablet made this possible? (2 marks)

Exercise 4 — Computers and the environment

1. What is e-waste? (2 marks)
2. State three negative impacts of computers on the environment. (3 marks)
3. Why does a short product life cycle add to the environmental impact of computers? (2 marks)
4. Propose three environmentally responsible practices that can reduce the negative effects of computer use. (3 marks)
5. What useful item can be produced from disposed computer parts in a school project? (2 marks)

Exercise 5 — Mixed review + scenarios

1. Ridgeview JHS bought six devices to support students living with disabilities. Name the general term for such equipment and the two families it falls under. (3 marks)
2. [Review] The school records offices scan student ID cards to enter attendance. What category of device reads the barcode directly into the computer? (2 marks)
3. [Review] Give one reason those offices should store data on a Solid-State Drive rather than a Hard Disk Drive. (2 marks)
4. A new teacher uses a screen reader herself. Which disability does this most likely support? (1 mark)
5. State two effects computer laboratories have on electricity bills if machines are left on all day, and one control practice. (2 marks)

Answers / marking scheme

Exercise 1

1. Assistive technology is a device or equipment used to increase, maintain or improve the functional abilities of persons with disabilities. (2 marks — "device" 1, "improve abilities of persons with disabilities" 1)

2. Any three: hearing aid; screen reader; Braille printer/keyboard; prosthetic limb; wheelchair. (1 mark each)
3. It makes sounds louder for a person who is hard of hearing. (2 marks)
4. A Braille printer (Braille embosser). (1 mark)
5. He can stand, walk and move about by himself, so he regains independence and can attend school like other learners. (2 marks)

Exercise 2

1. Adaptive technology is existing technology or everyday equipment modified/specially arranged to suit the way a person with a disability works. (2 marks)
2. Any two: picture-based communication app on a tablet; strap-on spoon/fork handle; raised-dot (Braille-adapted) keyboard. (1 mark each)
3. A picture-based communication (AAC) app; classified as adaptive technology because an ordinary tablet has been adjusted to fit the user. (2 marks)
4. Hearing aid — Assistive; strap-on handle — Adaptive; screen reader — Assistive; raised-dot keyboard — Adaptive. (1 mark each)
5. It allows learners with different disabilities to learn in the same classroom with their mates, using tools adjusted to each learner. (2 marks)

Exercise 3

1. A small, light computer that can easily be carried and used in different places. (2 marks)
2. Laptop, tablet, smartphone, notebook/e-reader. (½ mark each)
3. Any two: portability (use anywhere); rechargeable battery operation; compact size; supports multitasking. (1 mark each)
4. Any two: limited battery life; smaller screen/keyboard; easier to lose or steal; weaker performance compared with desktops. (1 mark each)
5. Portability (carried to the aunt's house) and wireless communication/Internet (sent to the class group). (2 marks — 1 each)

Exercise 4

1. E-waste is discarded electronic equipment, such as dead computers and monitors, which pollutes soil and water. (2 marks)
2. Any three: high power consumption; e-waste pollution; short product life cycle; heat from machines; chemicals from manufacturing. (1 mark each)
3. Devices replaced and thrown away quickly pile up as e-waste, so more waste and pollution per short period. (2 marks)
4. Any three: switch off/sleep unused computers; repair and reuse instead of dumping; recycle through certified e-waste channels; buy energy-saving devices; print less. (1 mark each)
5. A working component such as a media player or storage unit assembled from old parts (e.g. old hard drive + monitor). (2 marks)

Exercise 5

1. Such equipment is called assistive and adaptive technology; the two families are assistive (adds an ability) and adaptive (adjusts tools to fit the user). (3 marks)
2. A barcode reader/scanner, which is a direct data entry device. (2 marks)
3. It has no moving parts, so the valuable attendance records are safe from shock damage, and it stores/opens files quickly. (2 marks)
4. Visual impairment (she cannot see the screen well). (1 mark)
5. Impacts: high power consumption (bigger bills) and heat build-up on the machines. Practice: switch computers off at break and close of day, or enable sleep mode. (2 marks)

Next lesson: Health and Safety in the use of ICT tools — workstations, eye strain and posture (this paper tests it in P1 Q38 and P2 Q5(d)).